



Solution to an equation

1 Which of these best describes what a solution is? Tick **1** correct answer

- ☐ The value of the expressions on both sides of the equals sign when the same.
- ☐ The value of the expressions of both sides when the variable is zero.
- ☐ The value of the variable which makes both sides of the equation equal.
- ☐ The value of the variable which makes both sides equal to zero.

2 Which of these equations have a solution when $x = 4$? Tick **2** correct answers

- ☐ $4x = 4$
- ☐ $x + 4 = 0$
- ☐ $2x + 1 = 9$
- ☐ $3x - 15 = 3$
- ☐ $7 - 2x = -1$

3 Which of these is a solution to the equation $3a + 5 = 2a + 2$? Tick **1** correct answer

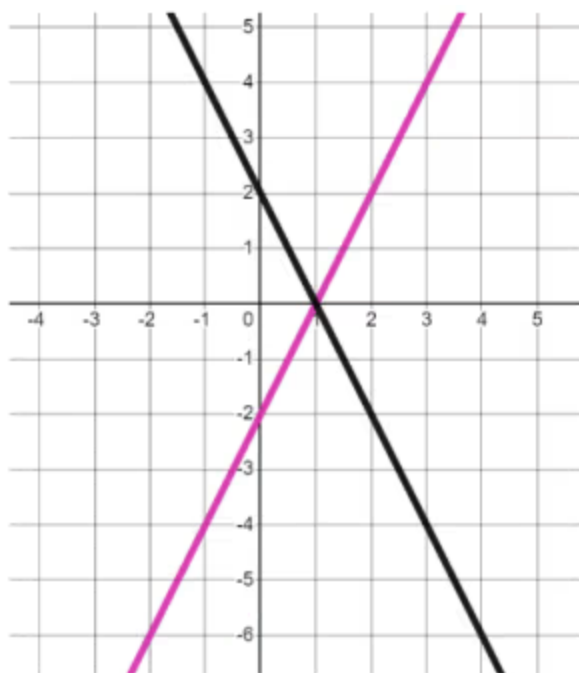
- ☐ $a = -5$
- ☐ $a = -3$
- ☐ $a = 0$
- ☐ $a = 2$
- ☐ $a = 4$

4 Which of these equations have no solution? Tick **2** correct answers

- ☐ $2x + 4 = 3x + 4$
- ☐ $x + 5 = 10 - x$
- ☐ $5x + 1 = 5x + 15$
- ☐ $2(x + 4) = 3x + 5$
- ☐ $4(2x + 1) = 8x$

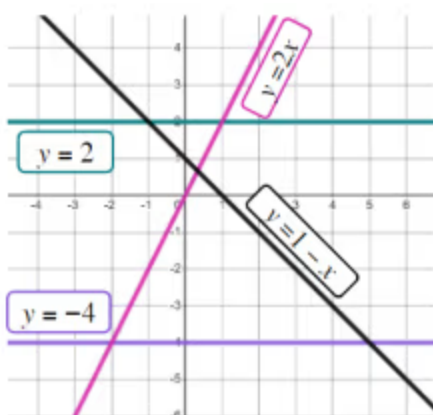


- 5 The graph shows the lines with equations $y = 2 - 2x$ and $y = 2x - 2$. What is the solution to the equation $2x - 2 = 2 - 2x$? Tick **1** correct answer



- ☐ $x = 2$
☐ $x = 1$
☐ $x = 0$
☐ $x = -2$

- 6 Use the graph to match each equation to its solution. Write the correct letter in each box



a	$1 - x = 2$
b	$1 - x = -4$
c	$2x = 2$
d	$2x = -4$

	solution when $x = 5$
	solution when $x = -1$
	solution when $x = -2$
	solution when $x = 1$

